

EMPLOYEE TABLE:

Emp ID	FNAME	LNAME	Salary	Dept ID
1	Vihan	Kapoor	60000	101
2	Ananya	Shetty	35000	106
3	Riya	Guptha	45000	120
4	Aditya	Iyer	55000	135
5	Aman	khan	200000	101

DEPARTMENT TABLE:

Dept ID	Dept Name	Location
1	HR	HUBLI
2	Sales	BELGAUM
3	IT	BANGLORE

Relational Algebra Expressions

1. Employees in department ID 101

$\pi_{FNAME, LNAME}(\sigma_{DeptID=101}(EMPLOYEE))$

FNAME	LNAME
Vihan	Kapoor
Aman	khan

2. Employees with salary > 50000

$\pi_{FNAME, Salary}(\sigma_{Salary>50000}(EMPLOYEE))$

FNAME	Salary
Vihan	60000
Aditya	55000
Aman	200000

3. Department name located in Bengaluru

$\pi_{\text{DeptName}}(\sigma_{\text{Location}=\text{"BANGLORE"}}(\text{DEPARTMENT}))$

Dept Name
IT

4. Employees in dept ID:106 and with salary > 30000

$\pi_{\text{FNAME}, \text{Salary}}(\sigma_{\text{DeptID}=106 \text{ AND } \text{Salary}>30000}(\text{EMPLOYEE}))$

FNAME	Salary
Ananya	35000

5. Employees with salary < 40000

$\pi_{\text{EmpID}, \text{FNAME}}(\sigma_{\text{Salary}<40000}(\text{EMPLOYEE}))$

Emp ID	FNAME
106	Ananya